ECGR 5411 - Assignment 10

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Problem 1

Consider the system with:

$$F(s) = \frac{s-4}{(s-1)(s+2)}, \quad H(s) = \frac{1}{s+1}.$$

Using the Nyquist plot, determine whether the closed loop system with the transfer function $\frac{F(s)}{1+F(s)H(s)}$ is stable.

***Solution*:**

```
clc;clear;close all;
s = tf('s');

F1 = (s-4)/((s-1)*(s+2));
H1 = 1 / (s+1);
L1 = F1*H1 % open-loop tf
```

```
L1 =

         s - 4
  -------------------
  s^3 + 2 s^2 - s - 2

Continuous-time transfer function.
Model Properties
```

```
sys1 = F1/(1+F1*H1) % closed-loop tf
```

```
sys1 =

   s^4 - 2 s^3 - 9 s^2 + 2 s + 8
  ------------------------------
  s^5 + 3 s^4 - 10 s^2 - 6 s + 12

Continuous-time transfer function.
Model Properties
```

```
Lpole1 = pole(L1) % det. # of open-loop RHP poles
```

```
Lpole1 = 3x1
    1.0000
```

```
-2.0000
-1.0000
```

$\implies$ 1 RHP pole in the open-loop transfer function at $s = 1 \implies P = 1$.

```
% [re, im, w] = nyquist(L);
nyquist(L1)
title('\textbf{Nyquist Plot, Problem 1}','Interpreter','latex','FontSize',20);
subtitle('$L(s) = F(s)H(s)$','Interpreter','latex','FontSize',16);
xlabel('Real Axis','Interpreter','latex','FontSize',12);
ylabel('Imaginary Axis','Interpreter','latex','FontSize',12);
```

Note: The following Nyquist plot was created in MATLAB using the `nyquist()` function (Fig. 1), and then recreated in LaTeX using the `pgfplots` package (Fig. 2)!

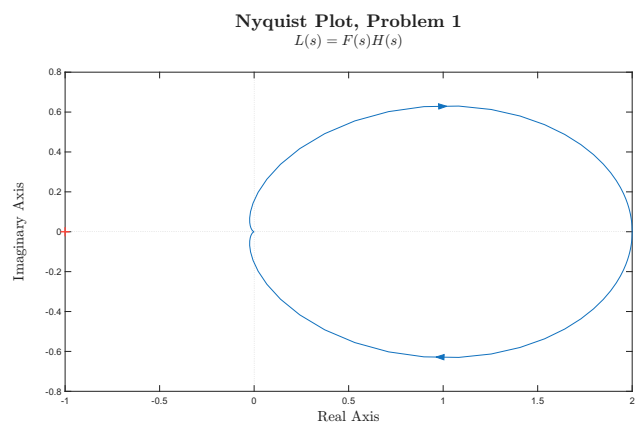


Figure 1: MATLAB Nyquist Plot, Problem 1

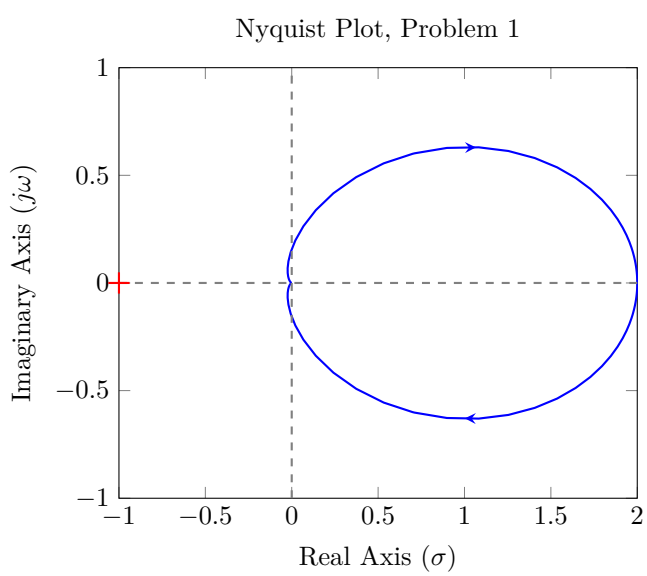


Figure 2: `pgfplots` Nyquist Plot, Problem 1

The parametric curve does not encircle $s = -1 \implies Z = N + P = (0) + (1) = 1$.

$\therefore Z = 1 \implies$ there is $\boxed{\text{one RHP zero}}$ in the closed-loop transfer function $\implies$ the system is $\boxed{\text{unstable.}}$

The following block was used to export the MATLAB Nyquist plot as a .csv file to be rebuilt in `pgfplots`.

```
% reshape table variables:
% re = squeeze(re);
% im = squeeze(im);
% w  = squeeze(w);
% create and export table:
% T = table(re, im, w, 'VariableNames', {'Re','Im','omega'});
% writetable(T, 'nyquist_data3.csv'); grid on;
```

Validated using closed-loop transfer function pole locations:

```
poles = pole(sys1);
disp(poles);
```

```
   -2.0000 + 0.0000i
   -1.6701 + 1.2990i
   -1.6701 - 1.2990i
    1.3403 + 0.0000i
    1.0000 + 0.0000i
```

There are two RHP poles in the closed-loop transfer function, which confirms that this system is unstable.

Problem 2

Consider the system with:

$$F(s) = \frac{1}{s^2 + 2s + 2}, \quad H(s) = \frac{1}{s+1}.$$

Using the Nyquist plot, determine whether the closed loop system with the transfer function $\frac{F(s)}{1+F(s)H(s)}$ is stable.

```
s = tf('s');
F2 = 1/(s^2+2*s+2);
H2 = 1 / (s+1);
L2 = F2*H2 % open-loop tf
```

```
L2 =

            1
  ---------------------
  s^3 + 3 s^2 + 4 s + 2

Continuous-time transfer function.
Model Properties
```

```
sys2 = F2/(1+F2*H2) % closed-loop tf
```

```
sys2 =

           s^3 + 3 s^2 + 4 s + 2
  ----------------------------------------
```

```
  s^5 + 5 s^4 + 12 s^3 + 17 s^2 + 14 s + 6

Continuous-time transfer function.
Model Properties
```

```
Lpole2 = pole(L2) % det # of open-loop RHP poles
```

```
Lpole2 = 3x1 complex
  -1.0000 + 1.0000i
  -1.0000 - 1.0000i
  -1.0000 + 0.0000i
```

$\implies$ No RHP poles in the open-loop transfer function $\implies P = 0$.

```
% [re, im, w] = nyquist(L);
nyquist(L2)
title('\textbf{Nyquist Plot, Problem 2}','Interpreter','latex','FontSize',20);
subtitle('$L(s) = F(s)H(s)$','Interpreter','latex','FontSize',16);
xlabel('Real Axis','Interpreter','latex','FontSize',12);
ylabel('Imaginary Axis','Interpreter','latex','FontSize',12);
```

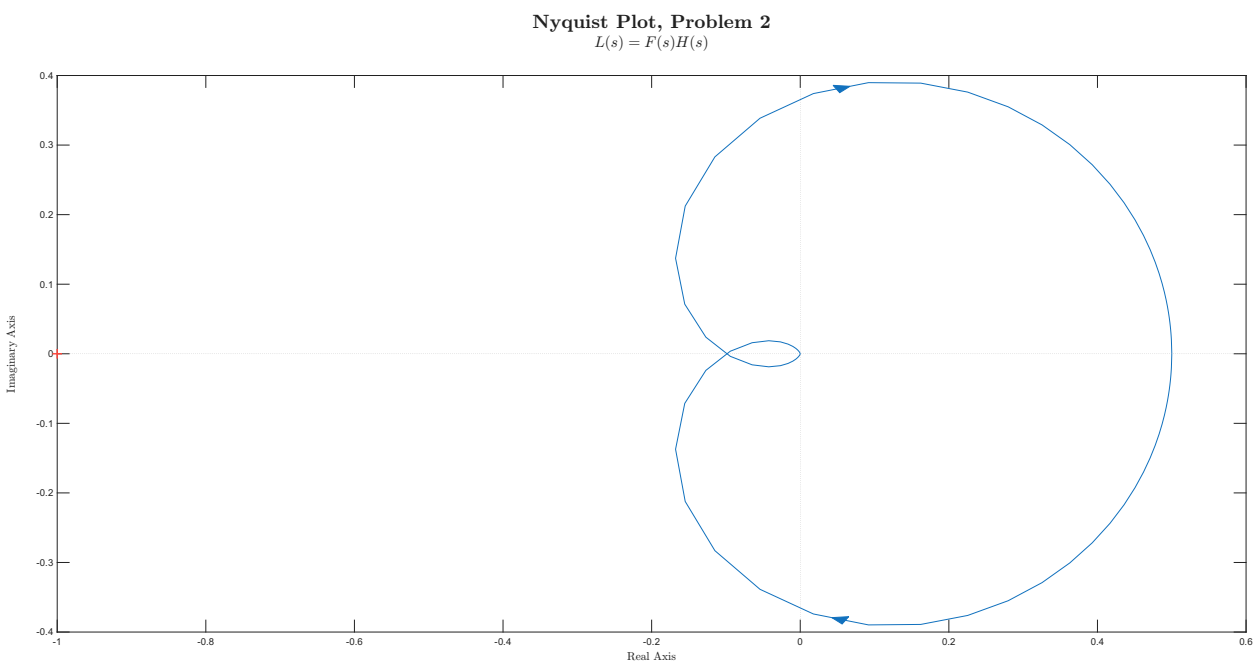


Figure 3: Nyquist Plot, Problem 2

Clearly, the Nyquist plot does not encircle $s = -1 \implies N = 0$. Thus,

$$Z = N + P = (0) + (0) = 0.$$

$\therefore Z = 0 \implies$ $\boxed{\text{no RHP zeros}}$ in closed-loop transfer function $\implies$ this system is $\boxed{\text{stable}}$.

Validate using closed-loop transfer function pole locations:

```
poles = pole(sys2);
disp(poles);
```

```
-1.6823 + 0.0000i
-0.6588 + 1.1615i
-0.6588 - 1.1615i
-1.0000 + 1.0000i
-1.0000 - 1.0000i
```

No RHP poles in the closed-loop transfer function confirm that this system is stable.

Problem 3

Suppose we add a gain K in the forward path, which makes the closed-loop transfer function equal to

$$\frac{F(s)}{\frac{1}{K} + F(s)H(s)}.$$

(a) Determine the range of values for K for which the system in Problem 1 is stable.

(b) Determine the range of values for K for which the system in Problem 2 is stable.

3(a)

Recall from the transfer function in Problem 1, where $K = 1$:

```
K = 1;
disp(Lpole1);
```

```
    1.0000
   -2.0000
   -1.0000
```

There is 1 RHP pole in the open-loop transfer function at $s = 1$.

The Nyquist plots in Figs. 1 and 2 show that the system is unstable when $K = 1$. By the Nyquist stability criterion, this system is stable when the parametric curve encircles $s = -1/K$ to have $Z = N + P = 0$. Thus, the parametric curve boundaries in Figs. 1 and 2 require:

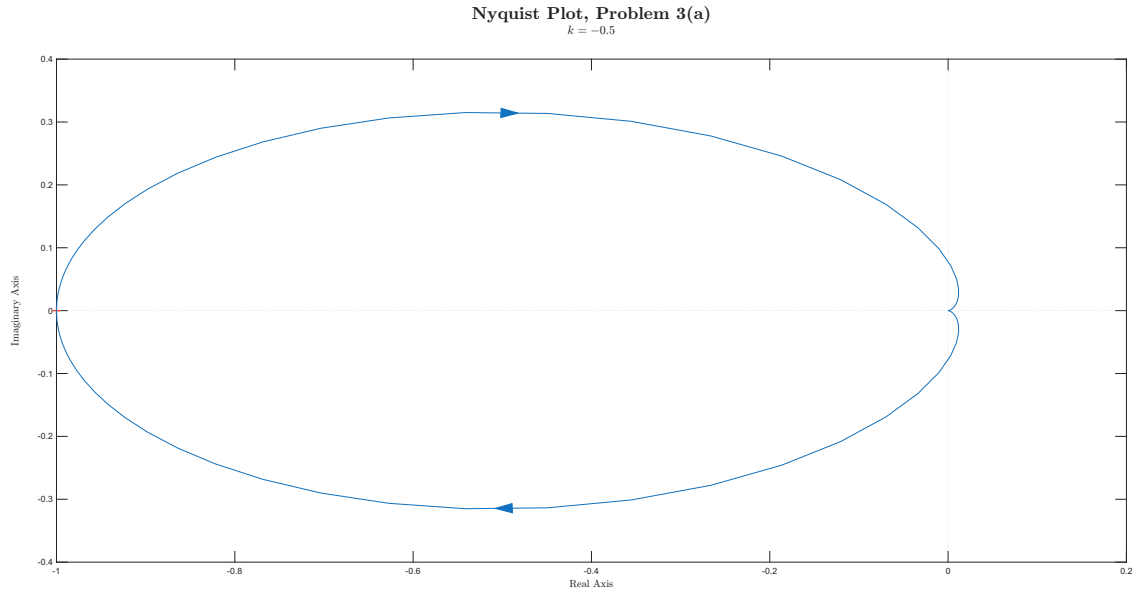
$$0 \leq -1/K \leq 2$$
$$\implies -\infty \leq K \leq -0.5$$

Therefore, $\boxed{K < 0.5}$ is the condition that guaranties stability.

This is verified in the following three Nyquist plots for increasing values of $|K|$.

```
% K = -0.5
nyquist(-0.5*L1);
title('\textbf{Nyquist Plot, Problem 3(a)}','Interpreter','latex','FontSize',20);
subtitle('$k=-0.5$','Interpreter','latex','FontSize',14);
xlabel('Real Axis','Interpreter','latex','FontSize',12);ylabel('Imaginary Axis','Interpreter
    ','latex','FontSize',12);
```

$K = -0.5$:

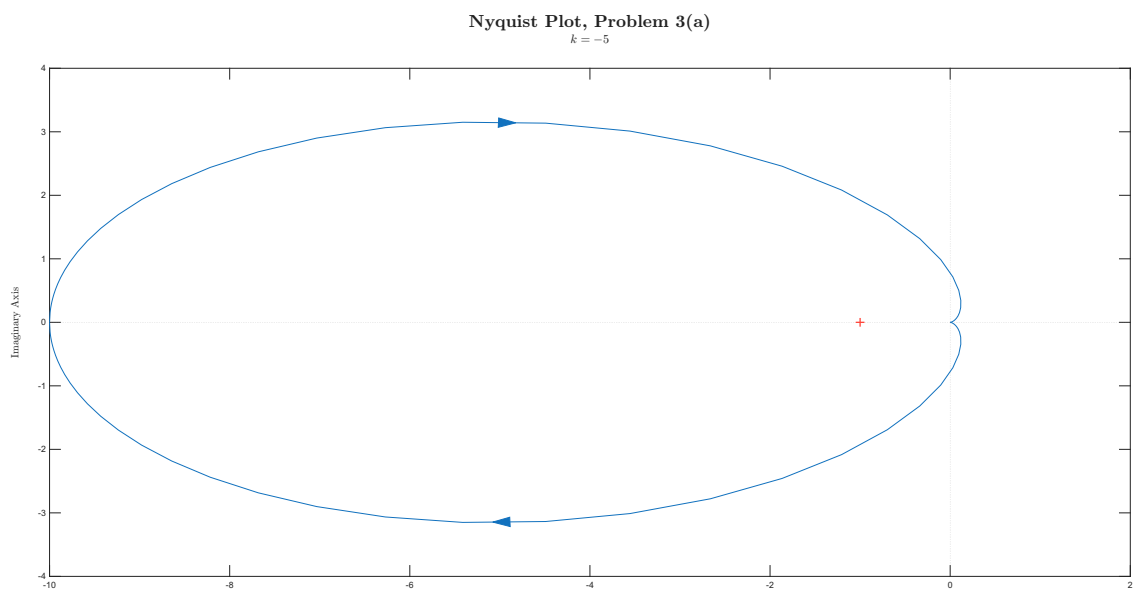


This Nyquist plot clearly encircles $s = -1$ on the edge of the parametric curve (apologies for the red plus not being very clear; however, it is in fact there when zoomed in).

Next, K was increased by a factor of 10:

```
% K = -5
nyquist(-5*L1);
title('\textbf{Nyquist Plot, Problem 3(a)}','Interpreter','latex','FontSize',20);
subtitle('$k=-5$','Interpreter','latex','FontSize',14);
xlabel('Real Axis','Interpreter','latex','FontSize',12);
ylabel('Imaginary Axis','Interpreter','latex','FontSize',12);
```

$K = -5$:



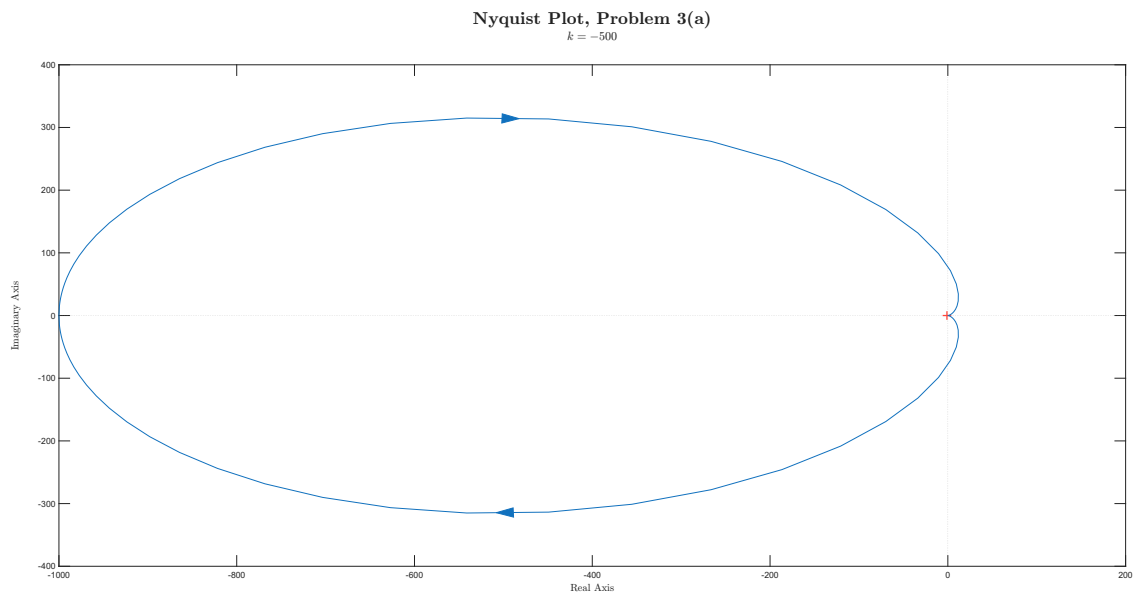
The red plus at $s = -1$ is much clearer in this Nyquist plot, and is obviously enclosed within the parametric

curve.

Finally, K was increased by an additional factor of 100:

```
% K = -500
nyquist(-500*L1);
title('\textbf{Nyquist Plot, Problem 3(a)}','Interpreter','latex','FontSize',20);subtitle('
    $k=-500$','Interpreter','latex','FontSize',14);
xlabel('Real Axis','Interpreter','latex','FontSize',12);
ylabel('Imaginary Axis','Interpreter','latex','FontSize',12);
```

$K = -500$:



Notice that as K becomes large, the location of $s = -1$ moves from the left to the right of the parametric curve in the Nyquist plot. This sufficiently defines the range for encircling $s = -1$ to achieve $Z = N + P = 0$ for stabilizing the system.

3(b)

Recall the transfer function in Problem 2, where $K = 1$:

```
disp(Lpole2); %
```

```
-1.0000 + 1.0000i
-1.0000 - 1.0000i
-1.0000 + 0.0000i
```

There are no RHP poles in the open-loop transfer function, so $P = 0$. This system is therefore stable when $s = -1/K$ is *outside* of the parametric curve (i.e. not encircled) so that $Z = N + P = 0$. Thus the boundaries of the curve in Fig. 3 require:

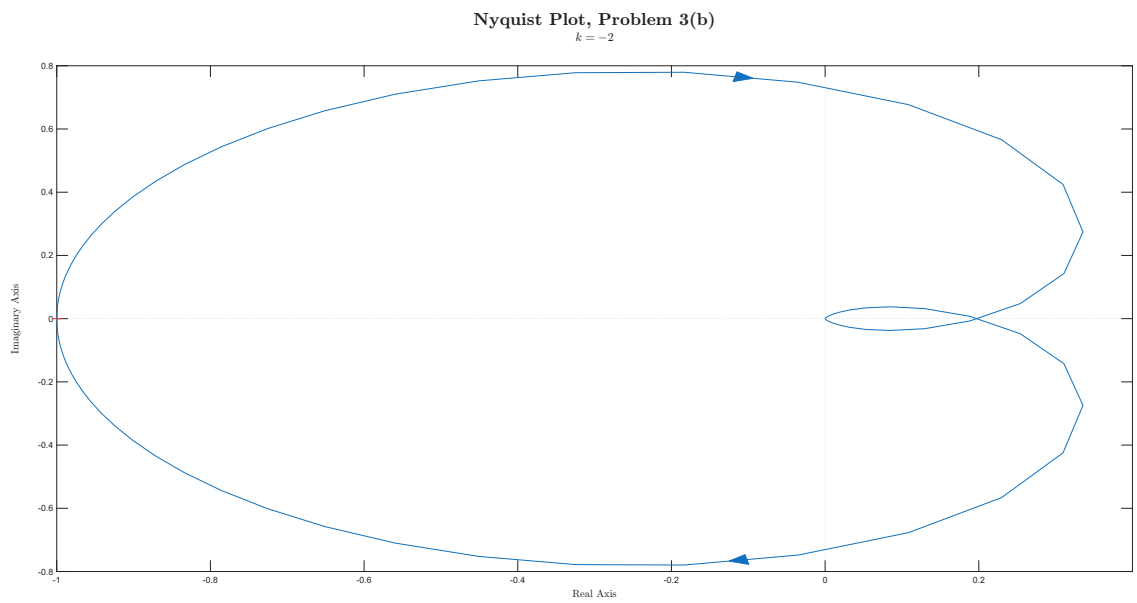
$$-0.1 < -1/K < 0.5$$
$$\implies -2 < K < 10$$

Therefore, $\boxed{-2 < K < 10}$ is the range of K values that guarantee stability.

This is verified in the following Nyquist plots, showing the resulting curves at the boundaries.

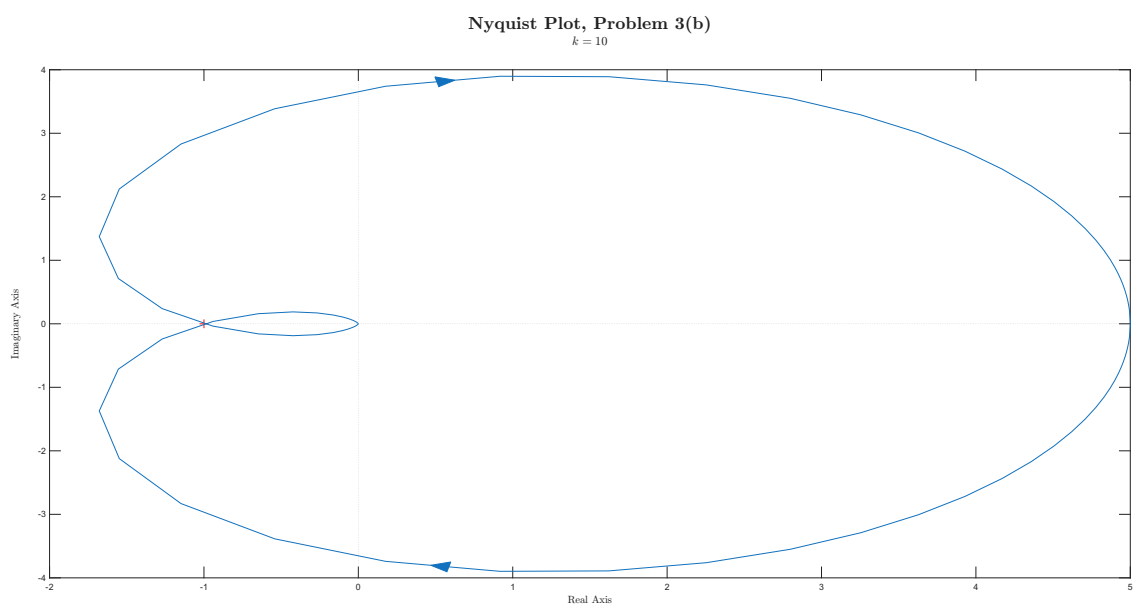
```
nyquist(-2*L2);
title('\textbf{Nyquist Plot, Problem 3(b)}','Interpreter','latex','FontSize',20);
subtitle('$k=-2$','Interpreter','latex','FontSize',14);
xlabel('Real Axis','Interpreter','latex','FontSize',12);
ylabel('Imaginary Axis','Interpreter','latex','FontSize',12);
```

$K = -2$:




```
nyquist(10*L2);
title('\textbf{Nyquist Plot, Problem 3(b)}','Interpreter','latex','FontSize',20);
subtitle('$k=10$','Interpreter','latex','FontSize',14);
xlabel('Real Axis','Interpreter','latex','FontSize',12);
ylabel('Imaginary Axis','Interpreter','latex','FontSize',12);
```

$K = 10$:



Although it requires zooming in at $s = -1$, both of these plots do, in fact, encircle this critical point, validating this range of stable K values.

Problem 4

Consider a second order system $F(s) = \frac{20}{s(0.5s+1)}$ and $H(s) = 1$.

(a) Compute the gain and phase margins, as well as the crossover frequencies, for $L(s) = F(s)H(s)$.

(b) Design a phase-lead controller to ensure a phase margin of at least 45°.

4(a)

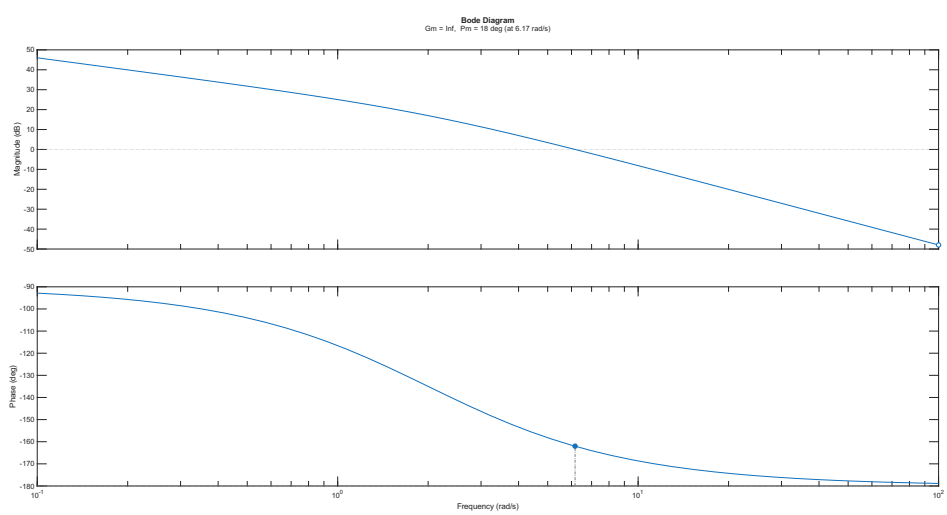
```
clc;clear;close all;
s = tf('s');
F4 = 20 / (s*(0.5*s+1))
```

```
F4 =

      20
  -----------
  0.5 s^2 + s

Continuous-time transfer function.
Model Properties
```

```
L4 = F4;
[Gm,Pm,Wcg,Wcp] = margin(L4)
```

```
Gm = Inf
Pm = 17.9642
Wcg = Inf
Wcp = 6.1685
```

```
margin(L4);
```



$\therefore$ Table 1 summarizes the results of the Bode plot above.

GM	PM	ω_{cg}	ω_{cp}
∞	$18°$	∞	6.17 rad/s

Table 1: Gain Margin, Phase Margin, and Crossover Frequencies for $L(s) = F(s)H(s)$

4(b)

The parameters chosen for the phase-lead controller are as follows:

- $\Phi_d = 45° - (18° + 5°) \approx 22°$ ($5°$ safety factor).
- ω_c chosen as 6 rad/s, near ω_{cp} in Table 1.

Then, the `syms solve()` function was used to choose α and τ that achieve the desired values above. K_c was manually adjusted to achieve the desired phase margin in the new Bode plot.

```
syms tau alpha;
assume(alpha<1);
wc = 1/sqrt(tau*alpha)
```

wc =

$$\frac{1}{\sqrt{\alpha\,\tau}}$$

```
phi_d = 45 - (Pm+5)
```

phi_d = 22.0358

```
phi_max = asin((1-alpha)/(1+alpha))*180/pi;

eq1 = wc == 6;
eq2 = phi_max == phi_d;

S = solve(eq1,eq2,[tau alpha]);
tauVal = double(S.tau)
```

tauVal = 0.0611

```
alphaVal = double(S.alpha)
```

alphaVal = 0.4543

```

K_c = 0.1305;
s = tf('s');
Cs = K_c * (tauVal*s+1)/(alphaVal*tauVal*s+1)
```

```
Cs =

  0.007978 s + 0.1305
  -------------------
     0.02778 s + 1
```

```
Continuous-time transfer function.
Model Properties
```

```
% validate with new bode plot;
[Gm1,Pm1,Wcg1,Wcp1] = margin(L4*Cs)
```

```
Gm1 = Inf
Pm1 = 50.0526
Wcg1 = Inf
Wcp1 = 1.9016
```

```
margin(L4*Cs);
title('\textbf{Bode Plot, Problem 4(b)}','Interpreter','latex','FontSize',20);
xlabel('$\omega$','Interpreter','latex','FontSize',12);
```

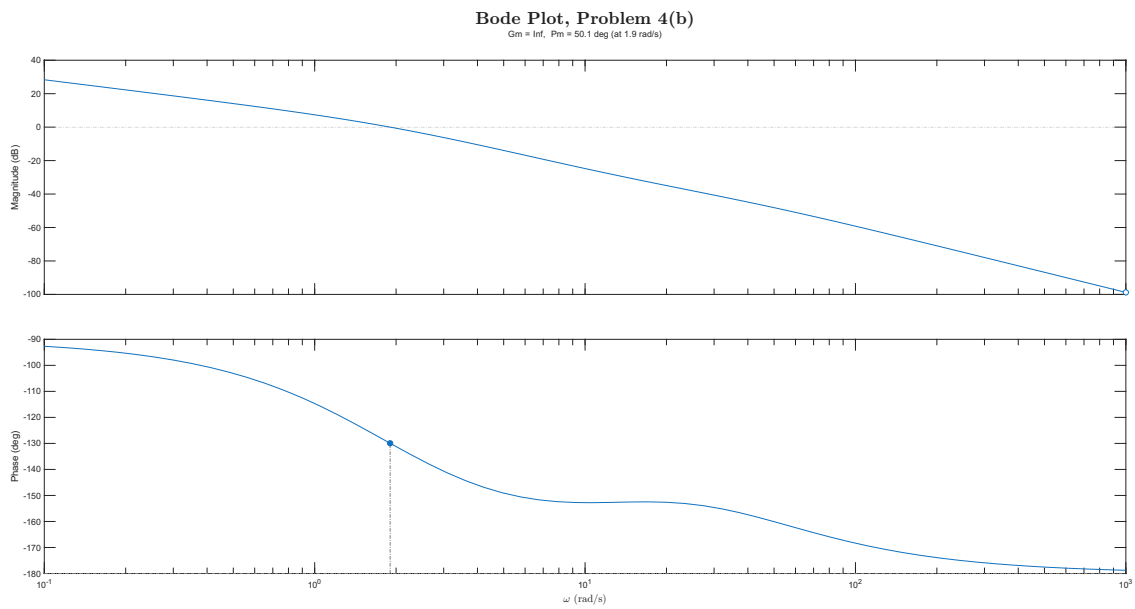


Table 2 summarizes the results for the Bode plot shown immediately above.

GM	PM	ω_{cg}	ω_{cp}
∞	$50.1°$	∞	1.9 rad/s

Table 2: Gain Margin, Phase Margin, and Crossover Frequencies for $C(s)L(s) = C(s)F(s)H(s)$

$\therefore$ The phase margin is sufficient for this design, at the cost of decreasing the bandwidth by nearly three times.